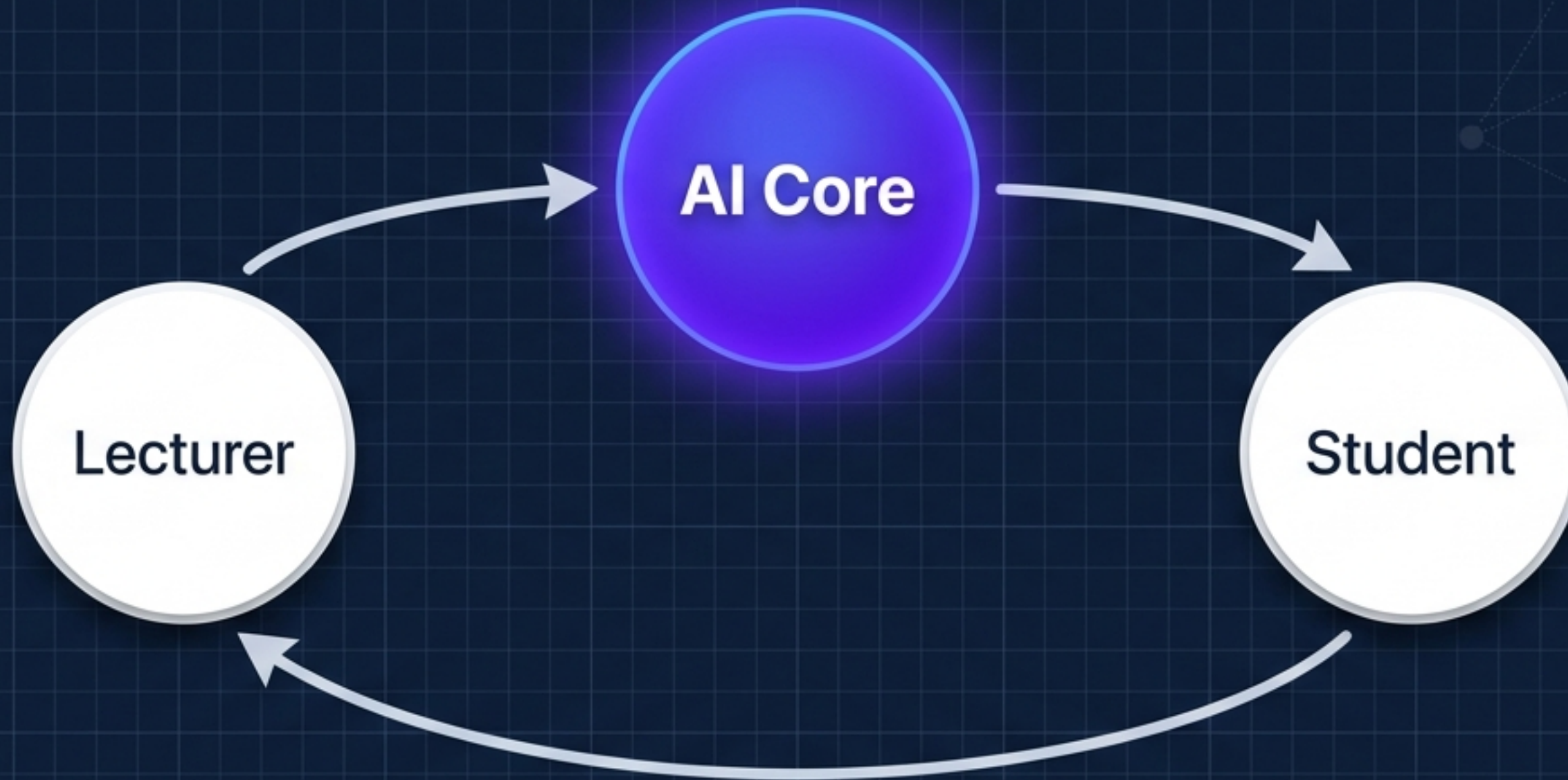




QuizGen: The Closed-Loop AI Practice Platform

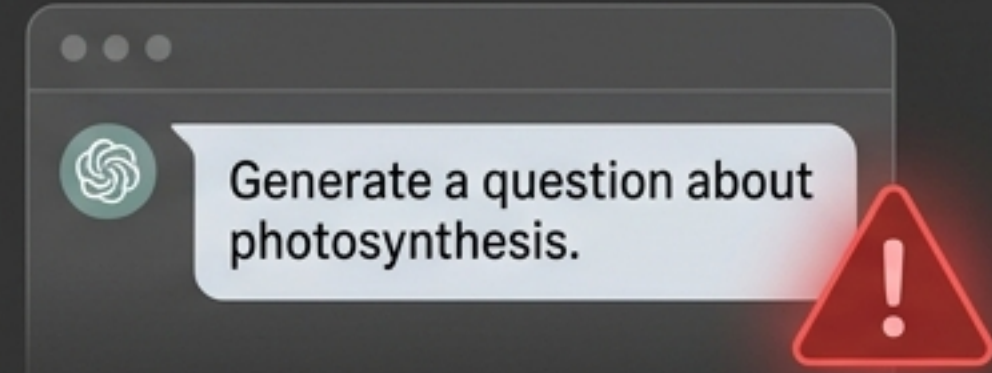
Pattern-Aware RAG-Based MCQ Generation from Lecture Materials.

The Manual Grind



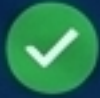
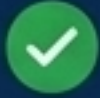
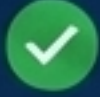
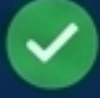
- High cognitive fatigue for educators
- Hours spent drafting stems and distractors
- Materials remain fragmented across platforms

The Raw LLM Hazard



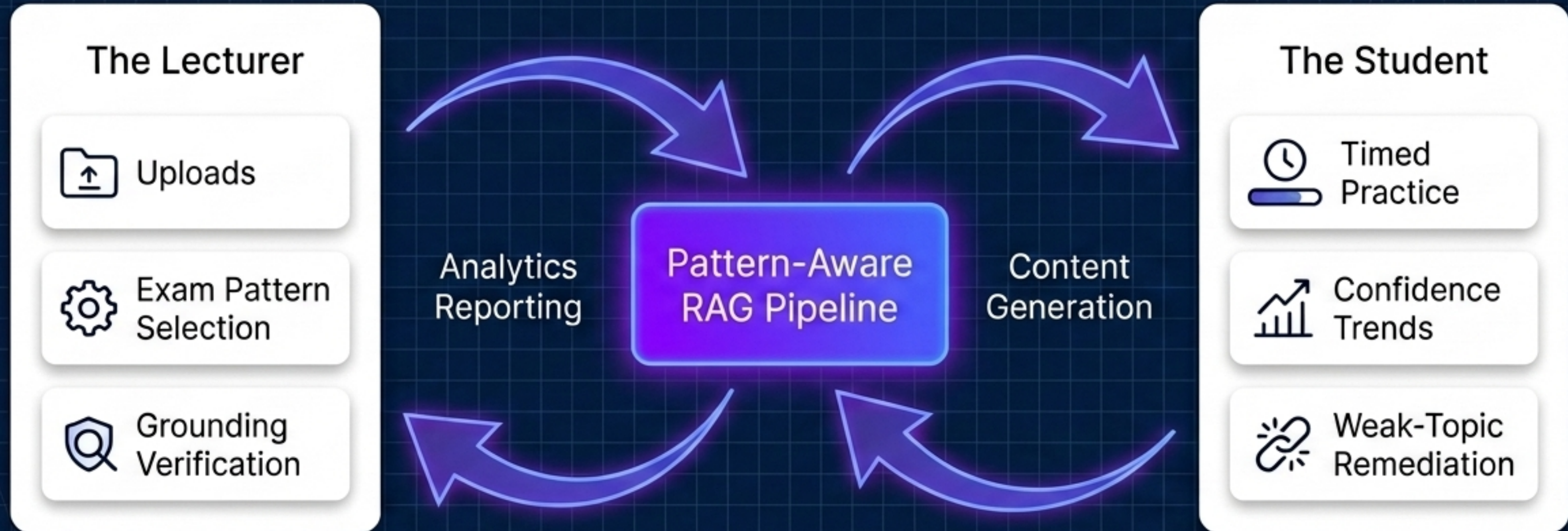
- Severe hallucination risks
- Out-of-syllabus drift
- Cognitive flatness (defaults to basic 'Remember' level)

The **educational bottleneck** requires a system that combines **AI generation speed** with **strict factual anchoring** and **pedagogical control**.

Feature	Generic ChatGPT	E-QGen	BUE AQG	Quillionz & Questgen	QuizGen
Source Grounding	×	×	×		
Exam Pattern Control	×		×	×	
Bloom Level Control	×	×	×		
Practice Workflow	×	×	×	×	
Performance Analytics	×	×	×	×	

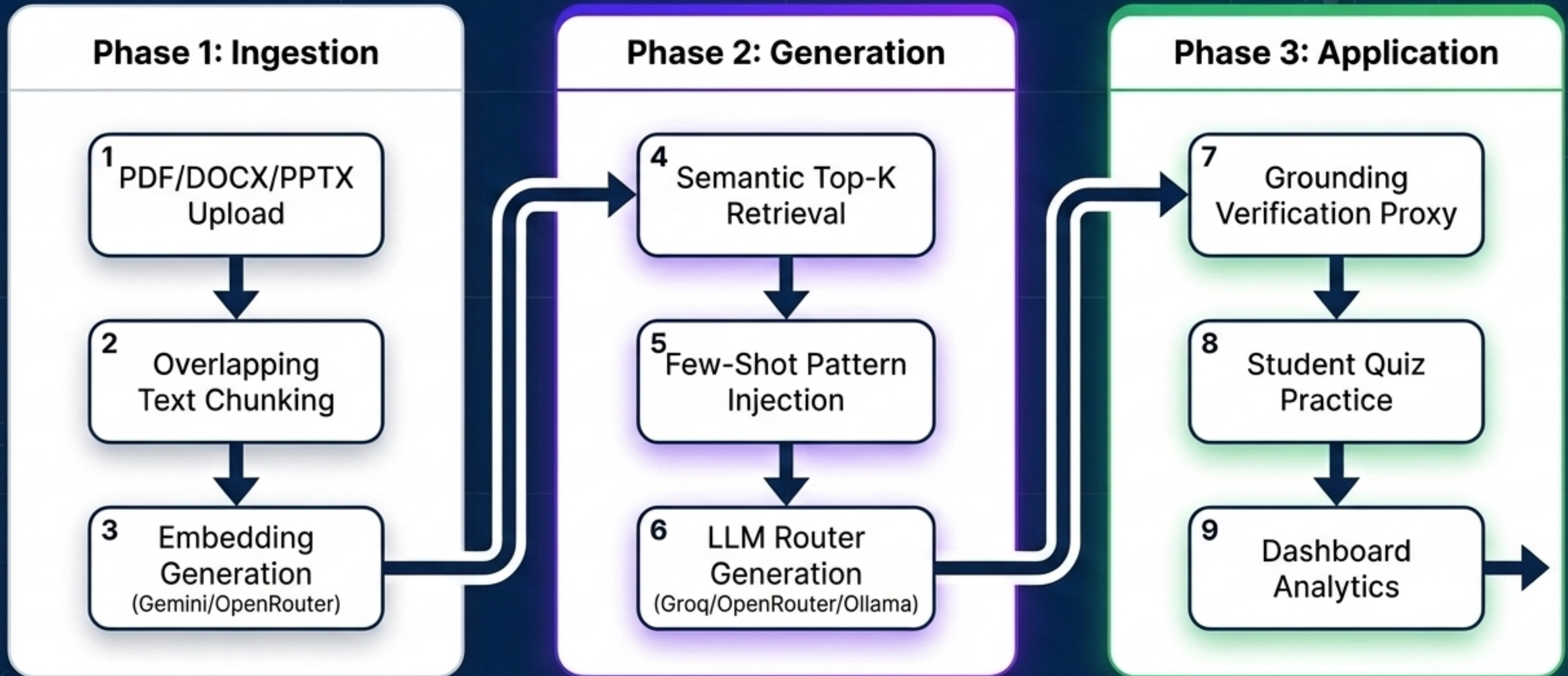
QuizGen integrates lecture ingestion, pattern-aware generation, grounding review, quiz practice, and topic-mastery analytics in one thesis prototype.

Closed-Loop Ecosystem

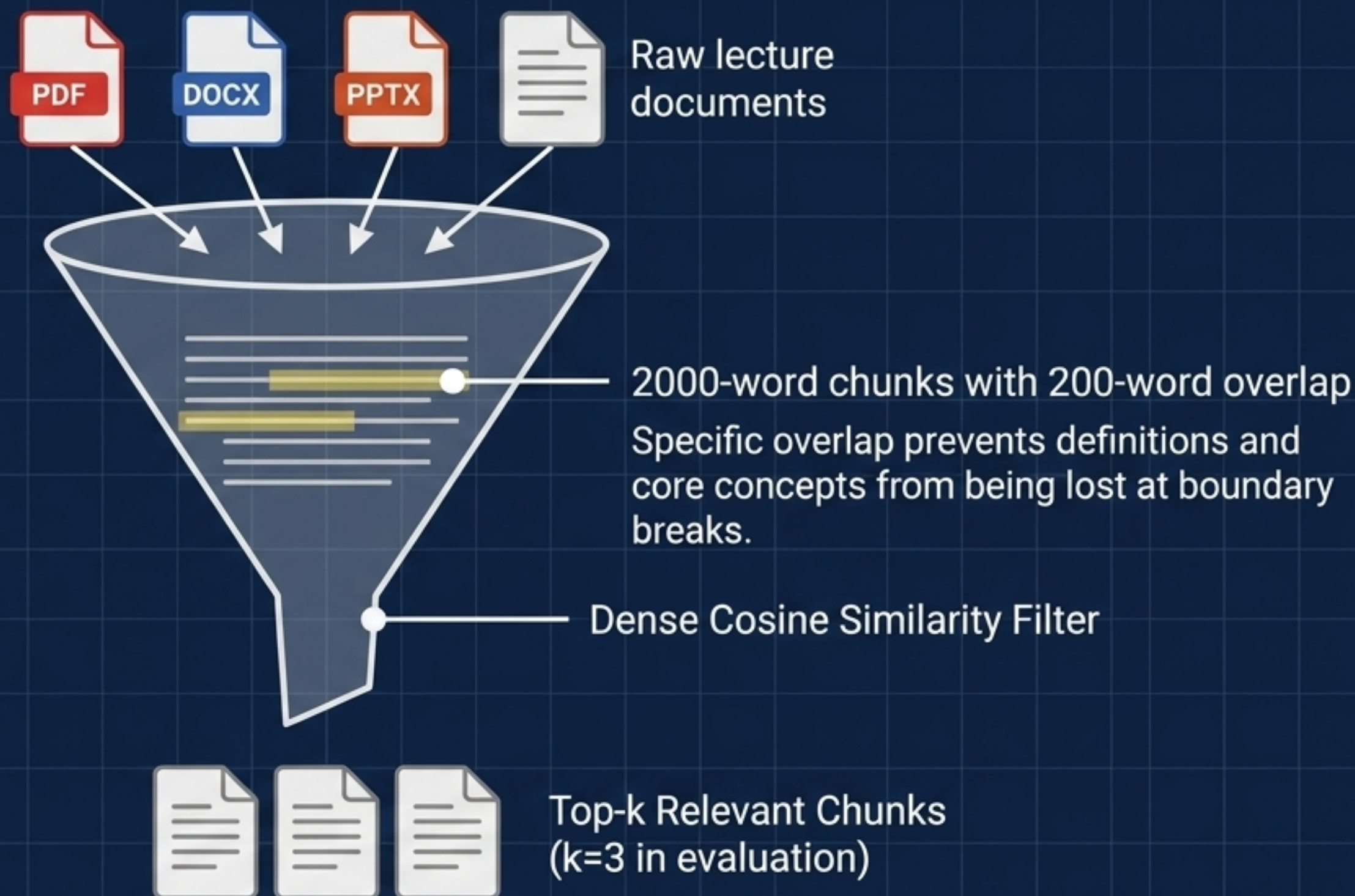


QuizGen connects generation, review, practice, and analytics in one locally hosted FastAPI/SQLite prototype.

The 9-Stage Pipeline



Knowledge Blueprint



By forcing the LLM to read a highly relevant, semantically selected 2000-word slice of the syllabus before answering, generation is steered toward course-specific facts.

Innovation 2 - Few-Shot Pattern Conditioning

[Top-K Retrieved
Chunks]

+

[Lecturer Exam
Pattern JSON]

+

[Bloom Difficulty
Sliders]

=

[Grounded MCQ
Prompt]

Pattern & difficulty

Optionally pin to an extracted exam pattern and shape Bloom-level distribution.

Exam pattern

Refresh list

No pattern

^

No pattern

✓

ISM Final

Difficulty distribution

40%

40%

20%

Drag the edges between colors (or use arrow keys when a handle is focused). Total stays 100%.

EASY

40

MEDIUM

40

HARD

20

Total: 100% - Locked to 100

The AI does not guess the format. It receives exact sample questions and strict distribution constraints, reduces the default "easy definition" bias.

Innovation 3 - The Grounding Proxy



A built-in mathematical heuristic **flags potentially weakly grounded MCQs**. It acts as an **automated review assistant**, focusing the lecturer's attention toward questions that may **deviate** from the vector embedding space of the lecture.

AI Drafts. Humans Verify.

Grounding Badges:
Proxy score in action to verify semantic alignment.

Review · Database Fundamenta... LLM check Export Practice →

10 MCGs 10/10 grounded completed

Database Fundamentals - Print Demo Save name

Grounding score 100%

Heuristic: questions with non-empty stem and answer. Use LLM check for evidence-based scoring as the document.

1 What is the primary purpose of a database management system? remember easy

A. A. To provide a user interface for applications B. B. To manage and control access to data

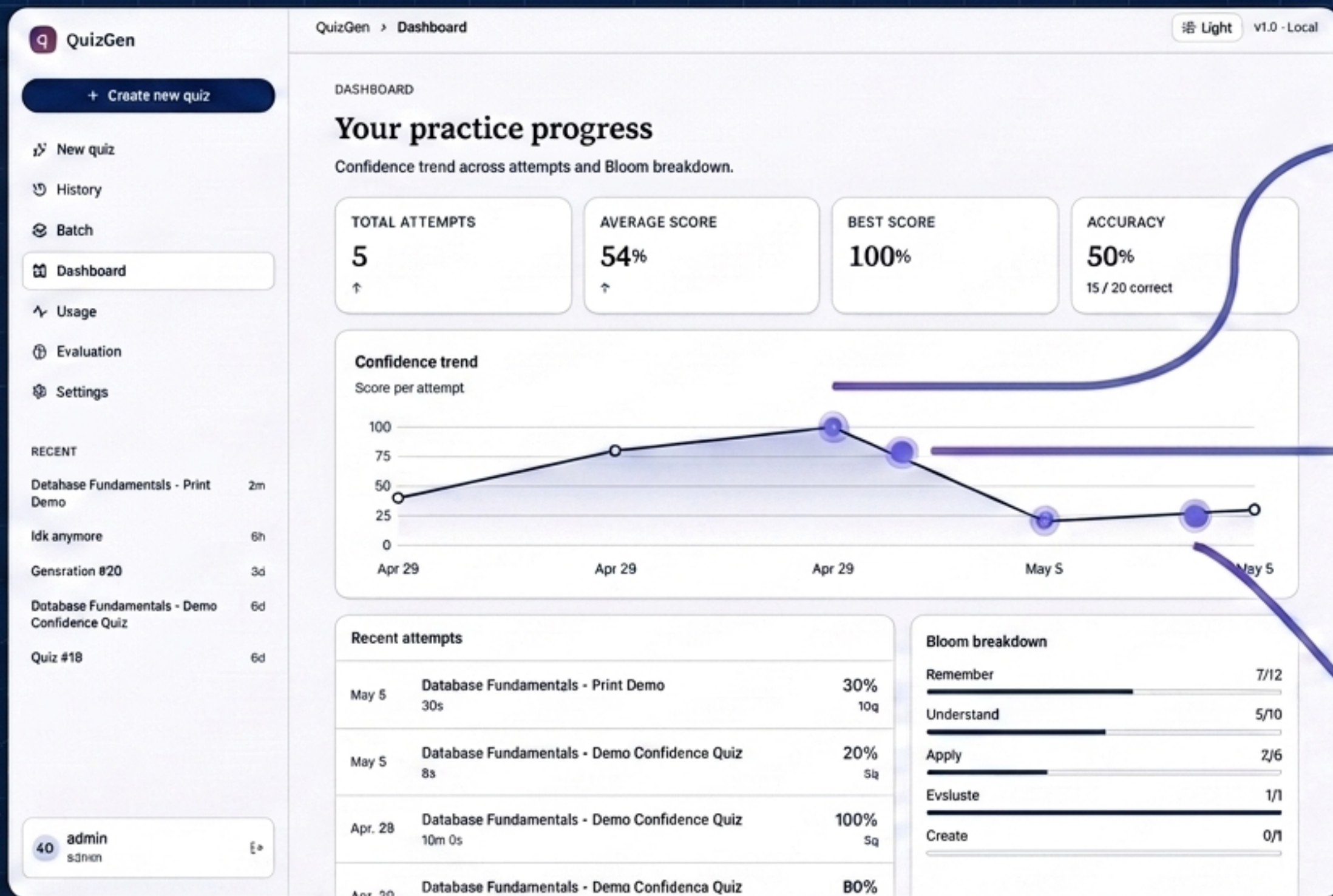
C. C. To perform calculations and data analysis D. D. To store and retrieve files

2 Which of the following is a characteristic of a relational database? remember easy

Human Review:
Lecturers inspect answer keys, explanations, Bloom labels, and grounding before reuse.

The review step prevents blind automation. The lecturer checks the generated structure, exports or regenerates if needed, and approves final use.

The Student View: Practice & Analytics



Chronological Tracking:
Attempt-based measurement of practice progress over time, replacing isolated quizzes.

Topic Mastery Isolation:
Identifies topics scoring below 70% and flags them as "weak".

Remedial Generation:
One-click generation of focused quizzes targeting weak topics from source material.

Experimental Setup: Data & Scope

Vanilla Prompting

Raw LLM. No retrieval.
No pattern.

RAG-Only

Adds Retrieval (Top-3).
No pattern.

Full System (QuizGen)

Adds Retrieval + Pattern
+ Bloom Control.

Scope: • 10 topics • 4 documents • 3 repeats • 6 questions per topic

Metrics Evaluated

↑ Semantic Grounding

↓ Bloom KL Divergence

↑ LLM Judge Score

Isolating variables proves which pipeline layer
solves specific AI flaws.

Results: Reducing Source Drift

Chart 1: Semantic Grounding

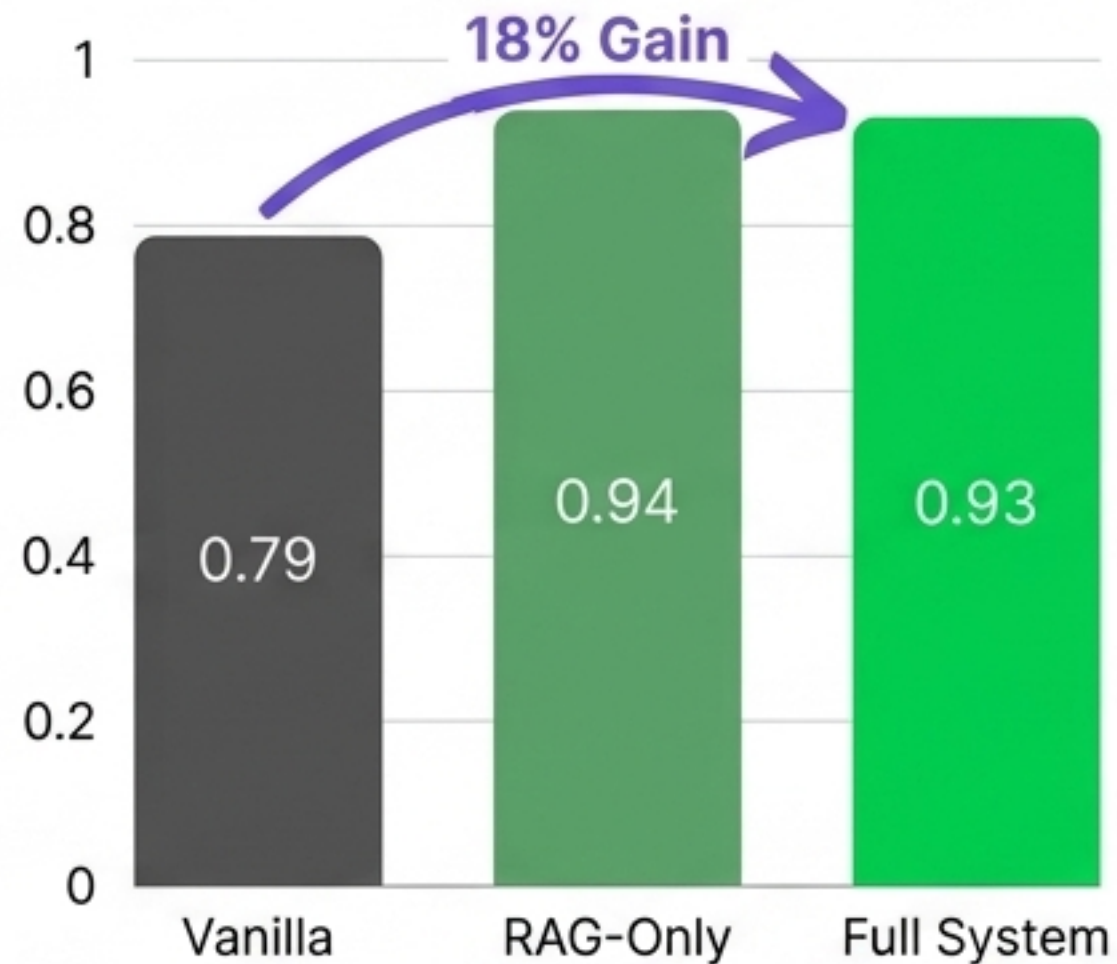
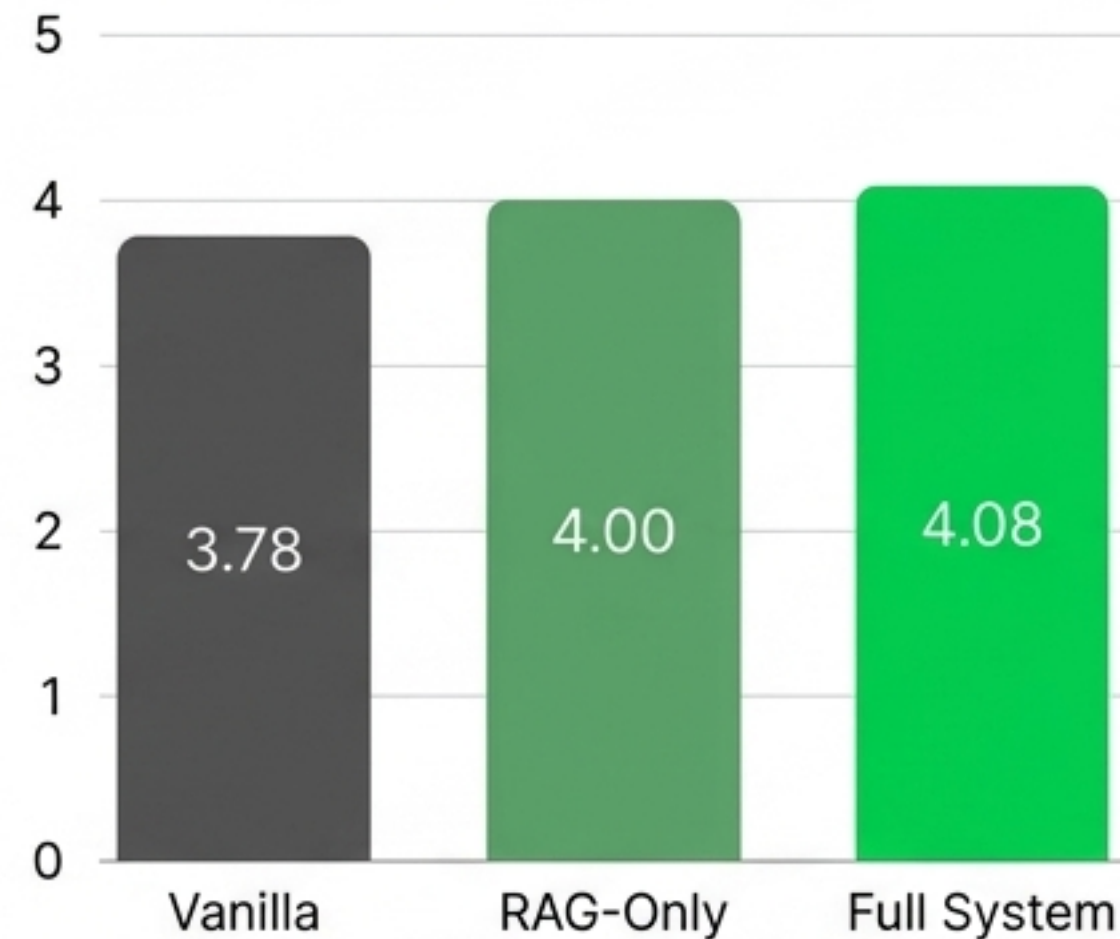
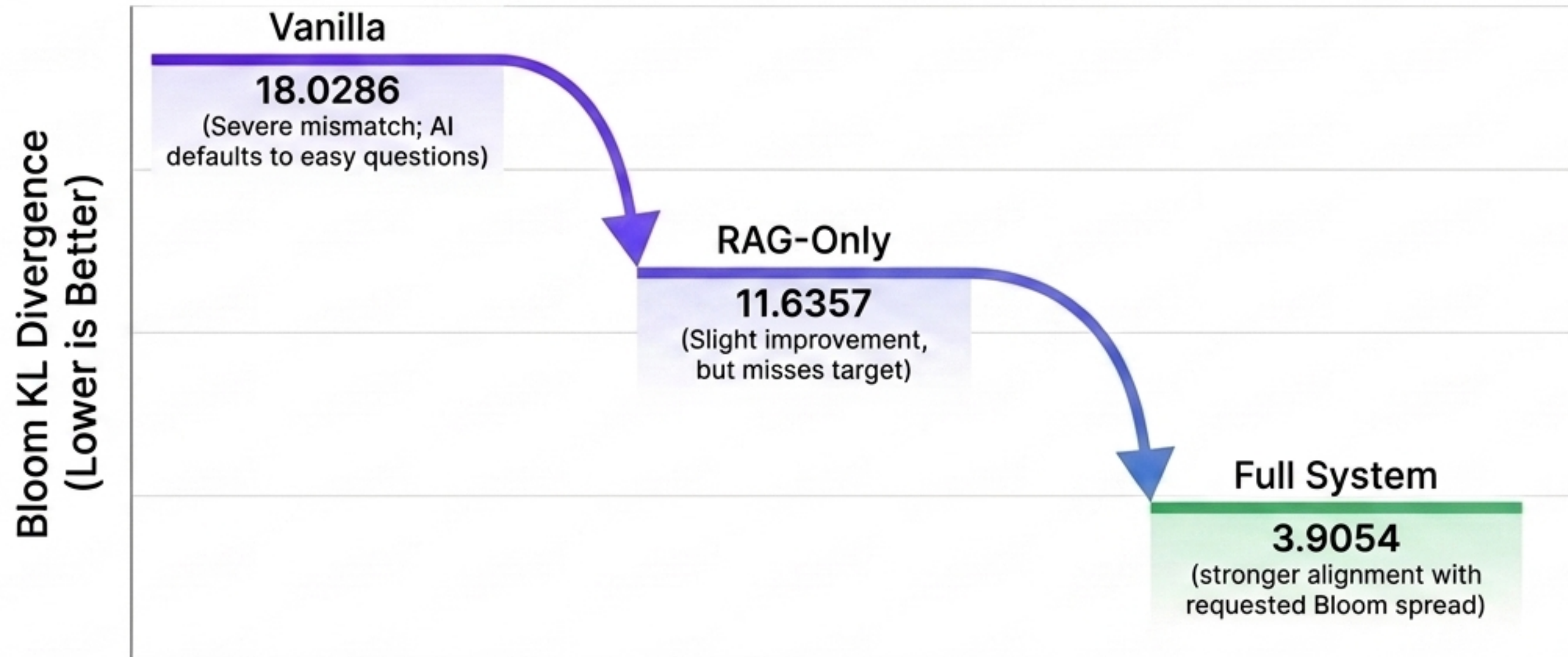


Chart 2: LLM Judge Score (out of 5)



Key Insight: The RAG pipeline substantially improves **semantic grounding** by steering generation toward retrieved lecture material.

Results: Controlling Cognitive Depth



RAG controls **WHAT** the AI asks about. Pattern conditioning controls **HOW** the AI asks. Only the Full System successfully guides the model to write more Apply and Analyze questions instead of just “Remember” definitions.

The Boundary of Automation (Failure Analysis)

Failure Analysis			
	Topic	Trigger	Value
1	cpu_scheduling	Bloom Mismatch	27.020
2	acid_transactions	Bloom mismatch or low judge/answer-verification risk	3.5

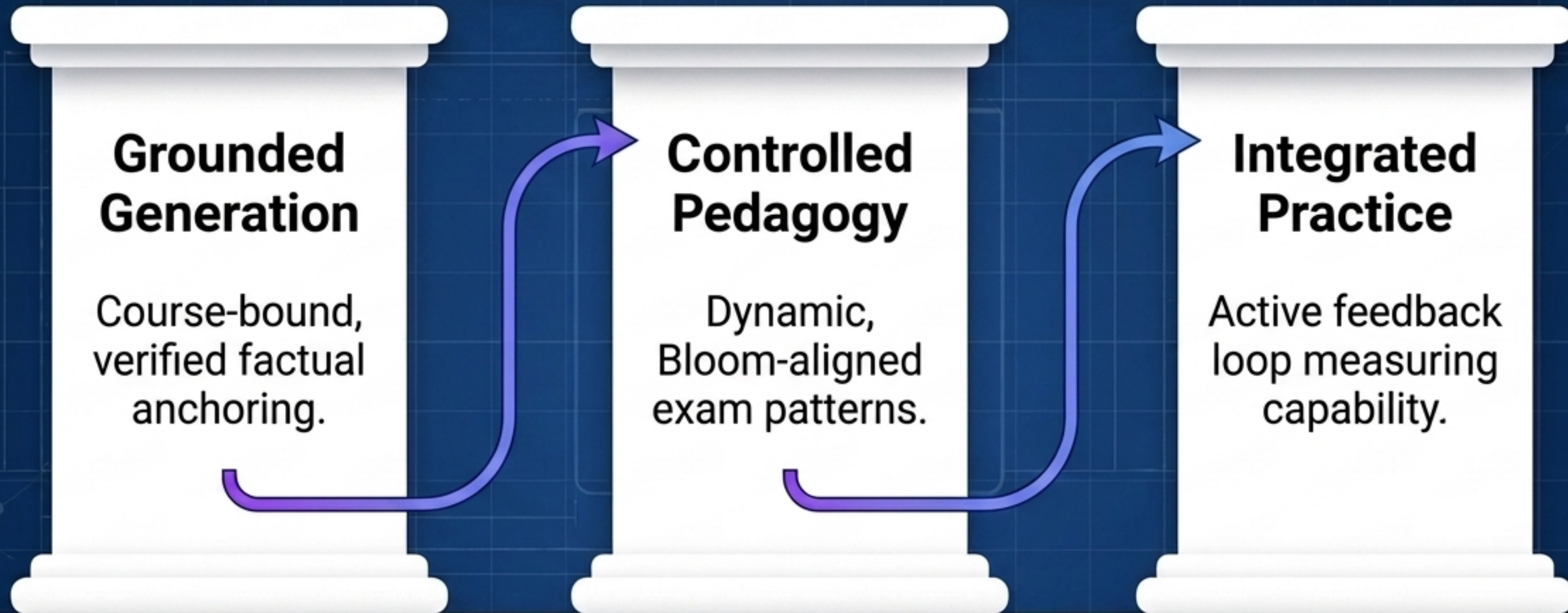
Semantic Closeness
to Source



Cognitive
Complexity

Insight: Complex topics still require human review. Some outputs show Bloom mismatch, topic confusion, or answer-verification risk, confirming the system is a drafting assistant rather than an autonomous examiner.

The Future of AI in Assessment



The future of educational AI lies in closed-loop systems weaving automated drafting, human validation, and targeted student practice.